

In 4 HW

1) 6

2) $\nabla AB = -4 = -\frac{3}{2}$

$\therefore \nabla \text{perpendicular} = \frac{2}{3}$

$y - y_1 = m(x - x_1)$

$y - 3 = \frac{2}{3}(x - 2)$

$y = \frac{2}{3}x + \frac{5}{3}$

3) $y' = 3x^2 - 2x$

$3x^2 - 2x + 1 = 0$

$(3x + 1)(x - 1) = 0$

$x = 1, y = -1 \quad (a)$

$x = -\frac{1}{3}, y = \frac{5}{3} \quad (d)$

4 i, iii, iv

5 $\tan \theta = \frac{1}{2}$

$\theta = 26.6, 206.6$

Consol

1 $\int \sqrt{5x+4} dx = \int u^{\frac{1}{2}} \cdot \frac{1}{5} du$

$= \frac{2}{3} u^{\frac{3}{2}} = \frac{2}{15} (5x+4)^{\frac{3}{2}} + C$

2 $\int (2e^{3x} + 8 \sin 5x) dx$
 $= \frac{2}{3} e^{3x} - \cos 5x + C$

3 a) $-\frac{1}{2} \cos(2x+1) + C$

b) $\frac{3}{2} e^{2x} + C$

c) $4e^{x+5} + C$

d) $-\frac{1}{2} \sinh(2x) + C$

e) $-\frac{1}{3} \sec^2 3x + C$

f) $\frac{1}{4} \sec 4x + C$

g) $-6 \cos(\frac{1}{2}x+1) + C$

h) $-\tan(2x) + C$

i) $\frac{1}{2} \csc(2x) + C$

j) $\frac{1}{3} (\sin 3x + \cos 3x) + C$

4 a) $\int e^{2x} - \frac{1}{2} \sinh(2x-1) dx$
 $= \frac{1}{2} e^{2x} + \frac{1}{4} \cosh(2x-1)$

b) $\int (e^x + 1)^2 dx$
 $= \int e^{2x} + 2e^x + 1 dx$
 $= \frac{1}{2} e^{2x} + e^x + x + C$

c) $\int (\sec^2 2x)(1 + \sinh 2x) dx$
 $= \int \sec^2 2x + \sec^2 2x \sinh 2x dx$
 $= \frac{1}{2} \tan 2x + \frac{1}{2} \sec 2x + C$

e) $\int \frac{e^{3-x} + \sinh(3-x) + \cosh(3-x)}{-e^{3-x} + \cosh(3-x) - \sinh(3-x)} dx$

5 $\int_{-1}^1 (3-2x)^4 dx = \left[\frac{1}{5} (3-2x)^5 \right]_{-1}^1$

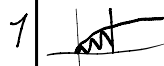
$= \frac{1}{5} - (-3125)$

$= 312.4$

6 $\int_0^2 e^{3x} dx$

$= \left[\frac{1}{3} e^{3x} \right]_0^2$

$= \frac{e^6}{3} - \frac{1}{3}$

7 

$\int_{-2}^1 (1+2x)^{\frac{1}{2}} dx$
 $= \left[\frac{2}{3} (1+2x)^{\frac{3}{2}} \right]_{-2}^1$

$= \frac{2}{3} \left[\frac{27}{2} - \frac{1}{2} \right]$

$= \frac{27}{3} = 9$

8 $\int_0^4 (2+x^5)^{10} dx$
 $= \int_0^4 (u^{10})^{\frac{1}{5}} du$

$= \int_0^4 u^2 du$

$= \frac{1}{3} u^3$

$= \frac{(2+x^5)^3}{3} + C$

b) $\int x^3 \cos(x^4) dx$
 $= \int \frac{1}{4} \cos u du$

$= \frac{1}{4} \sin u$

$= \frac{1}{4} \sin x^4 + C$

c) $\int \frac{4x}{2x^2+1} dx$

$= \int u^{-1} du$

$= \ln u$

$= \ln 2x^2+1 + C$

d) $\int e^x \sec^2(e^x) dx$
 $= \tan e^x + C$

e) $\int \frac{\sin x}{1-\cos x} dx$

$= \int \frac{\sin x}{u} \times \frac{1}{-\sin x} du$

$= \int -\frac{1}{u} du$

$= -\ln u$

$= -\ln(1-\cos x) + C$

f) $\int \cot x dx$

$= \int \frac{\cos x}{\sin x} dx$

$= \int \frac{1}{u} du$

$= \ln |\sin x| + C$

g) $\int \cos x e^{\sin x} dx$

$= \int e^u du$

$= e^{\sin x} + C$

h) $\int x^4 \sec x^{10} \tan x^{10} dx$

$= \int \frac{1}{10} \sec u \tan u du$

$= \frac{\sec u^{10}}{10} + C$

Exam Qs

$$\begin{aligned}
 1 \int_0^{\frac{\pi}{2}} \cos 3x \, dx \\
 &= \left[\frac{1}{3} \sin 3x \right]_0^{\frac{\pi}{2}} dx \\
 &= \frac{1}{3} \sin \frac{\pi}{2} \\
 &= \frac{\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 2 \int \frac{8x^2+4}{4x^3+6x-1} \, dx \\
 = \frac{2}{3} \ln(4x^3+6x-1) + C
 \end{aligned}$$

$$\begin{aligned}
 3 \int 5 \cos x \sin^3 x \, dx \\
 = \frac{9}{4} \sin^4 x + C
 \end{aligned}$$

$$\begin{aligned}
 4 \text{ i. } (e^x - 2)^2 - 1 &= 0 \\
 e^x &= 3 \\
 x &= \ln 3
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } f'(x) &= e^x \times (2e^x - 4) \\
 2e^x &= 4 \\
 e^x &= 2 \\
 x &= \ln 2 \\
 y &= -1
 \end{aligned}$$

$$\begin{aligned}
 \text{iii. } \int_0^{\ln 3} (e^x - 2)^2 - 1 \, dx \\
 &= \int_0^{\ln 3} e^{2x} - 2e^x - 3 \, dx \\
 &= \left[\frac{1}{2} e^{2x} - 4e^x + 4x \right]_0^{\ln 3} \\
 &= 9 - 12 + 4 \ln 3 \\
 &= 4 \ln 3 - 3
 \end{aligned}$$